

ReSpect Manual

Arsia Takeh and Sachin Shanbhag

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1 Introduction

ReSpect infers continuous and discrete relaxation spectra from dynamic moduli data obtained from small-angle oscillatory shear (SAOS) experiments. These experiments yield the frequency-dependent dynamic moduli, $G^*(\omega) = G'(\omega) + iG''(\omega)$. The relaxation spectra and complex modulus are mathematically linked by:

$$G'(\omega) = \int_{-\infty}^{\infty} \frac{\omega^2 \tau^2}{1 + \omega^2 \tau^2} h(\tau) d\ln \tau = \sum_{i=1}^N g_i \frac{\omega^2 \tau_i^2}{1 + \omega^2 \tau_i^2} \quad (1)$$

$$G''(\omega) = \int_{-\infty}^{\infty} \frac{\omega \tau}{1 + \omega^2 \tau^2} h(\tau) d\ln \tau = \sum_{i=1}^N g_i \frac{\omega \tau_i}{1 + \omega^2 \tau_i^2} \quad (2)$$

where $h(\tau)$ is the continuous relaxation spectrum (CRS), and the N modes $\{g_i, \tau_i\}$ describe the discrete relaxation spectrum (DRS).

Note: In the literature, $H(\tau)$ is often used to denote the CRS.¹

1.1 Source

The basic algorithms used in this paper are described in the paper:

- Takeh A, Shanbhag S: A Computer Program to Extract the Continuous and Discrete Relaxation Spectra from Dynamic Viscoelastic Measurements, Appl. Rheol. 23 (2013) 24628.

Please cite this paper if you use ReSpect and find it helpful.

2 Log of Changes

The last full release of ReSpect was in Feb 2013. In this version (v2.0), several changes have been made including bug-fixes, tweaks to the basic algorithm, and addition of new capability.

(i) Bug-Fixes

- Looks for repeated frequencies in input data, and removes duplicates. This often happens due to time-temperature superposition of data gathered at different frequencies (in `common/GetExpData.m`).
- In the GUI, the discrete spectrum was incorrectly plotted (in `gui/updateplot`). This is now fixed.
- The default plotting setting for `discSpec` have been changed to show the spectrum. Note that when the modes g_i are negative, you may see warnings about inability to represent negative numbers on a log-log plot.

(ii) Tweaks to the Algorithms

- The residual and smoothness constraints are now normalized by the number of data points in the input file n , and the number of grid-points for representing the CRS n_l , respectively.

¹The $H(\tau) = \log h(\tau)$ used in ReSpect has a different meaning.

- The method used to determine the optimal λ_c has been tweaked. We define a function based on the normalized $\rho - \eta$ curve,

$$\epsilon(\lambda) = \frac{\rho(\lambda)}{\max \rho(\lambda)} + \frac{\eta(\lambda)}{\max \eta(\lambda)},$$

and choose a λ_c that minimizes $\epsilon(\lambda)$. In general, the new criterion leads to somewhat larger values of λ_c , and hence smoother spectra.

(iii) New Features

- By default, ReSpect uses Douglas Schwarz's SuperSmoother program² to filter input data (in `common/GetExpData.m`). This is particularly handy when the input data is noisy.
- Introduced an independent directory Gt to infer $h(\tau)$ from $G(t)$ data via the equation:

$$G(t) = \int_{-\infty}^{\infty} h(\tau) e^{-t/\tau} d\ln\tau = \sum_{i=1}^N g_i e^{-t/\tau} \quad (3)$$

The original program to infer $h(\tau)$ from $G^*(\omega)$ is now housed in the subdirectory Gstar

- Introduced a directory structure to the programs

3 Program

ReSpect v2.0 contains two folders:

1. Gstar: Extract $h(\tau)$ from $G^*(\omega)$
2. Gt: Extract $h(\tau)$ from $G(t)$

Currently the two calculations are completely independent. Therefore, if the intended use is only to extract $h(\tau)$ from $G^*(\omega)$, then the Gt folder can be safely deleted.

ReSpect can be accessed either from:

- the command line
- a graphical user interface (GUI) (for Gstar with Matlab only)

To use ReSpect, the user needs to provide an **input file** which contains the $G^*(\omega)$ or $G(t)$ data, and **specify parameters** either by modifying the function file SetParameters, or using the GUI.

3.1 Using the Program

Depending on the type of calculation enter the appropriate sub-directory (Gstar or Gt).

²<http://www.mathworks.com/matlabcentral/fileexchange/17986-supersmoother/content/supsmu.m>

3.1.1 Interfaces

- *Command line*: By calling `contSpec` and `discSpec`, we get the CRS and DRS, respectively. Note that CRS has to be computed before DRS, since DRS is an approximation built on the CRS. Default settings can be adjusted by modifying the `SetParameters` function.
- *Graphical user interface (GUI)*: This interface is currently accessible only for `Gstar` in Matlab. The GUI can be opened by calling the function `specgui`. The advantage of the GUI is that all the adjustable parameters can be modified through the GUI, instead of the function `SetParameters`.

3.1.2 Input file

The experimental data $G^*(\omega)$ is reported at a set of discrete frequencies $\{\omega_i, G'(\omega_i), G''(\omega_i)\}$. It is saved as an ASCII text file³ with 3 columns. The first, second and third columns contain ω , $G'(\omega)$ and $G''(\omega)$ respectively. This text file is labeled as `Gst.dat` by default, although this can be changed if required through the `SetParameters` function. Thus, an example of `Gst.dat` looks like:

0.0370	12.4048	226.2025
0.0406	14.6341	246.8058
0.0455	17.2642	269.2831
0.0489	20.3672	293.8066
0.0536	24.0280	320.5642
0.0588	28.3462	349.7587
0.0645	33.4399	381.6140
0.0708	39.4517	416.3851
0.0777	46.5402	454.2931
0.0852	54.8993	495.6178
.	.	.
.	.	.
.	.	.

Similarly the experimental (or simulation) data $G(t)$ is reported as an ASCII text file with two columns, $\{t_i, G(t_i)\}$. This text file is labeled as `Gt.dat` by default.

3.1.3 Specifying Parameters

The parameters can be set either by modifying the `SetParameters.m` function file, or the GUI. We will first describe the parameters that appear in `SetParameters` file, and comment on the default settings. The GUI allows us to modify the default settings, without having to open the `SetParameters` file directly.

³a plain text file made with Notepad, for example

(i) **Overall parameters:**

- `verbose = 1`
Turns ON printing to screen and files (OFF = 0).
- `plotting = 1`
Turns ON plotting functions (OFF = 0).

(ii) **Continuous spectrum:**

- `ns = 100`
Number of grid points to represent the continuous spectrum.
- `lamC = 0`
Use the L-curve method to determine λ_c (If not 0, the value used as λ_c instead of using L-curve method). This allows the user to specify the level of smoothing on an absolute scale (compare with `SmFacLam`).
- `SmFacLam = 0`
Smoothing factor, indirect way of controlling λ_c , relative to the one inferred from the L-curve method. Set between -1 (lowest λ explored) and 1 (highest λ explored). Default setting ($= 0$) uses λ_c determined from the L-curve.
- `GstFile = "Gst.dat"`
Default name for experimental complex moduli data, 3 columns ASCII file contains $[\omega, G', G'']$.
- `FreqEnd = 1`
Range of τ in $H(\tau)$ extracted from $G^*(\omega)$. Strictly, if the ω ranges from $\omega_{\min}$ to $\omega_{\max}$, then the "trustable" range of τ is $[e^{\pi/2}/\omega_{\max}, e^{-\pi/2}/\omega_{\min}]$. However, the smoothness criterion imposed on $H(\tau)$ allows us to be less careful.

choice	range of τ
3	$[e^{\pi/2}/\omega_{\max}, e^{-\pi/2}/\omega_{\min}]$
2	$[1/\omega_{\max}, 1/\omega_{\min}]$
1	$[e^{-\pi/2}/\omega_{\max}, e^{\pi/2}/\omega_{\min}]$

(iii) **Discrete spectrum**

- `Nopt = 0`
Specify number of discrete modes (> 0). If set equal to 0, then an internal algorithm is used to determine it automatically.
- `prune = 1`
Avoid modes with -ve ($= 0$: don't care). This *tries* to suppress -ve modes.

- `BaseDistWt = 0.5`
Controls spacing of τ_i , set between 0 and 1. If set to 0 the spacing is completely determined by $H(\tau)$. If set to 1, τ_i are equi-spaced.
- `condWt = 0.5`
Controls relative importance of error and conditioning in determining N_{opt} , set between 0 and 1. If set to 0, the number of modes are completely determined by error. On the other hand, setting it to 1 determines the number of modes completely based on the condition number.

3.1.4 Output

Depending on whether the verbose and printing flags are turned on or off, some of the output will be directed to the screen. Regardless, all the necessary output files are stored in the subfolder `output`. This includes the spectrum (`H.dat` for CRS, and `dmodes.dat` for DRS), the $G^*(\omega)$ corresponding to the spectrum (`Gfit.dat` for CRS, and `Gfitd.dat` for DRS) as plain text files.

The L-curve corresponding to the CRS is stored in the `rho-eta.dat` as three columns, $\{\lambda_i, \rho_i, \eta_i\}$.

The file `Nopt.dat` contains three columns, the number of modes, the corresponding error, and condition number of the matrix problem.

4 Example

In this example, we demonstrate the extraction of the CRS and DRS from $G^*(\omega)$ data. As an example, we use an experimental sample on binary linear-linear blends which spans over seven decades in frequency from Watanabe et al ⁴. Experimental dynamic moduli data for this example is provided as the file (`Gst.dat`).

We first ensure we are in the appropriate directory (`Gstar/`) and start an Octave or Matlab session.

At the command line, call the CRS function:

```
>> H = contSpec();
```

If we call the function without any parameters, it reads the parameters from the function `SetParameters`. This is the preferred mode of operation.

The program takes a while to run (typically of the order of a minute), depending upon the number of datapoints in the input file, and the discretization level of the spectrum.

Output is written into separate files: spectrum data (`H.dat`), L-curve data $\rho - \eta$ (`rho-eta.dat`) and dynamic modulus from the corresponding CRS (`Gfit.dat`). All the output files are stored in the `Gstar/output/` subfolder. The contents of these files are plotted and explained in figures 1, 2, and 3.

⁴Watanabe H, Ishida S, Matsumiya Y, Inoue T: Test of full and partial tube dilation pictures in entangled blends of linear polyisoprenes, *Macromolecules* 37 (2004) 6619-6631

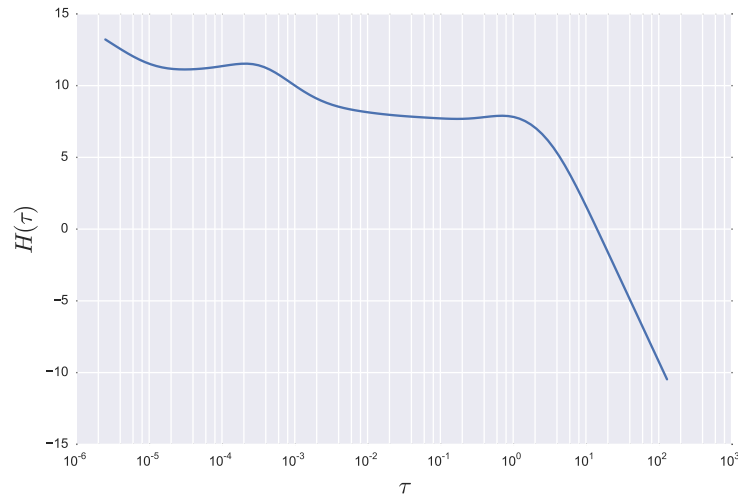


Figure 1: The extracted $H(\tau)$ (H.dat) on a semi-log plot. Note that the “actual” spectrum is $h(\tau) = e^{H(\tau)}$.

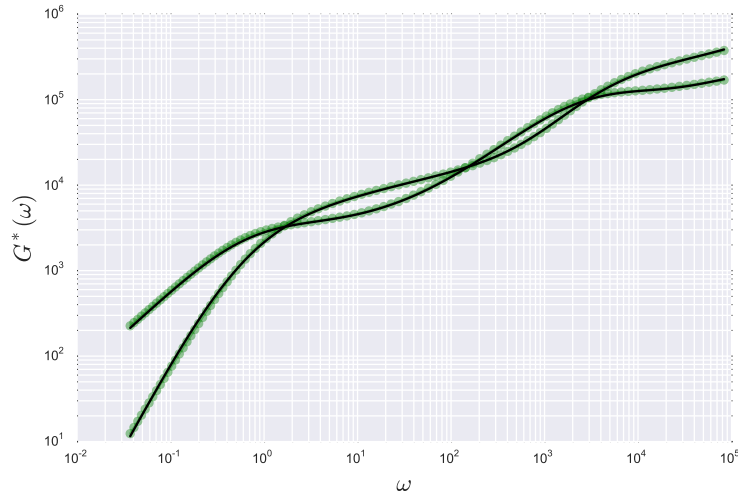


Figure 2: The circles represent the experimental $G'(\omega)$ and $G''(\omega)$ (Gst.dat). The solid lines represent the corresponding dynamic modulus from the corresponding CRS (Gfit.dat) in figure 1.

The $\rho - \eta$ curve is presented in figure 3. The location corresponding to the default λ_c chosen by the algorithm is also specially marked.

To compute the DRS, after the CRS has been determined, use the following command.

```
>> discSpec()
```

If we call the function without any parameters, it reads the parameters from the function SetParameters. This is the preferred mode of operation.

The program usually runs super-fast.

Outputs are written into separate files, spectrum data (dmodes.dat), the file which contains error and conditioning of the problem, used to determine the optimal number of modes (Nopt.dat) and dynamic

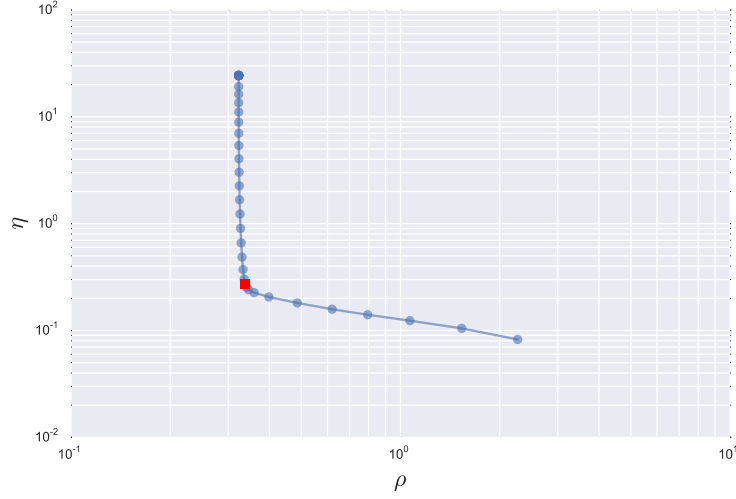


Figure 3: The $\rho - \eta$ curve (columns 2 and 3 of `rho-eta.dat`). The red square corresponds to the corner value of λ_c chosen by default.

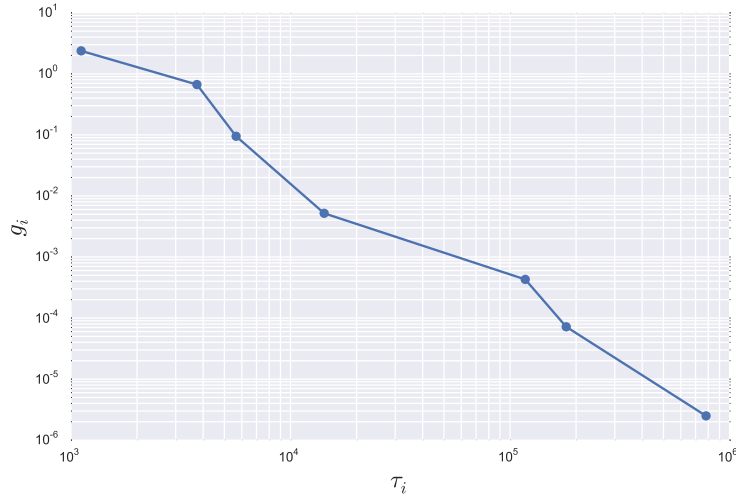


Figure 4: Discrete relaxation spectra with $N = 7$ modes. This data is contained in the file `dmodes.dat`. The three columns of this file contain the mode number, τ_i and g_i , in that order.

modulus from the corresponding DRS (`Gfitd.dat`). All the output files are stored in the `Gstar/output/` subfolder. The contents of these files are plotted and explained in figures 4, 5, and 6.

Using the default setting, the inferred DRS had 7 modes. The spectrum from file `dmodes.dat` is presented in figure 4. Note that sometimes negative modes cannot be avoided, and attempting to plot the DRS on a log-log plot might cause some warnings and complaints.

A comparison of the input data and the comparable $G^*(\omega)$ corresponding to the DRS is plotted in fig. 5.

Finally, figure 6 shows the contents of the `Nopt.dat` file, which is internally used to determine the optimal number of modes to use. The error between the computed $G^*(\omega)$ and the input data can be made smaller

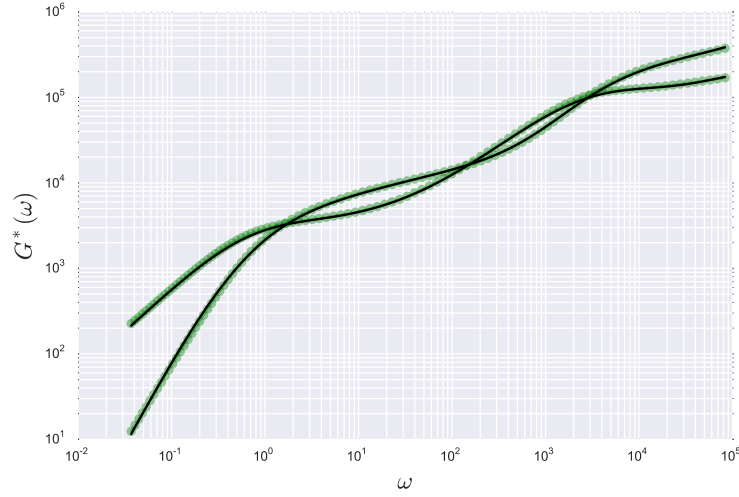


Figure 5: The circles represent the $G'(\omega)$ and $G''(\omega)$. The solid lines represent the corresponding dynamic modulus from the corresponding DRS.

by increasing the number of discrete modes in the DRS. However, this leads to a concurrent increase in the condition number of the underlying linear least squares problem. The goal is to pick an N_{opt} that is large enough, so that the error is small. This has to be balanced against choosing an N_{opt} that is small enough, so that the condition number is sufficiently small.

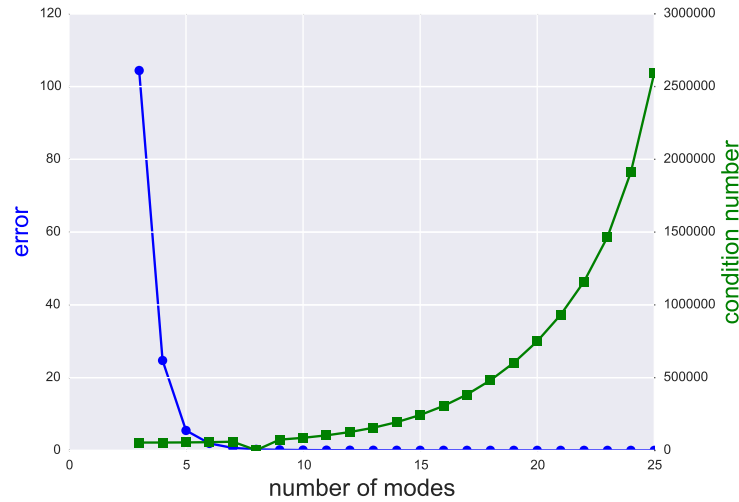


Figure 6: Plot to determine the optimum number of discrete modes. The default criterion used in the program yielded $N_{opt} = 7$. The relative error between the input data and the $G^*(\omega)$ inferred from the DRS is plotted on the left y-axis, while the condition number of the resulting linear least-squares problem is plotted on a logarithmic scale on the right y-axis.

Using the following command, we call GUI:

```
>> specgui()
```

Figure 7 depicts the GUI console. Several parameters, that may be adjusted through the `SetParameters` function, can alternatively be adjusted on using this panel.

The first thing is to decide whether we want to extract CRS or DRS. This selection can be made by using the radio button at top of the panel. Note that the CRS still has to be computed before the DRS. However, if you want to re-compute the DRS by changing some of the setting (number of modes), you can do so, without having to redo the CRS.

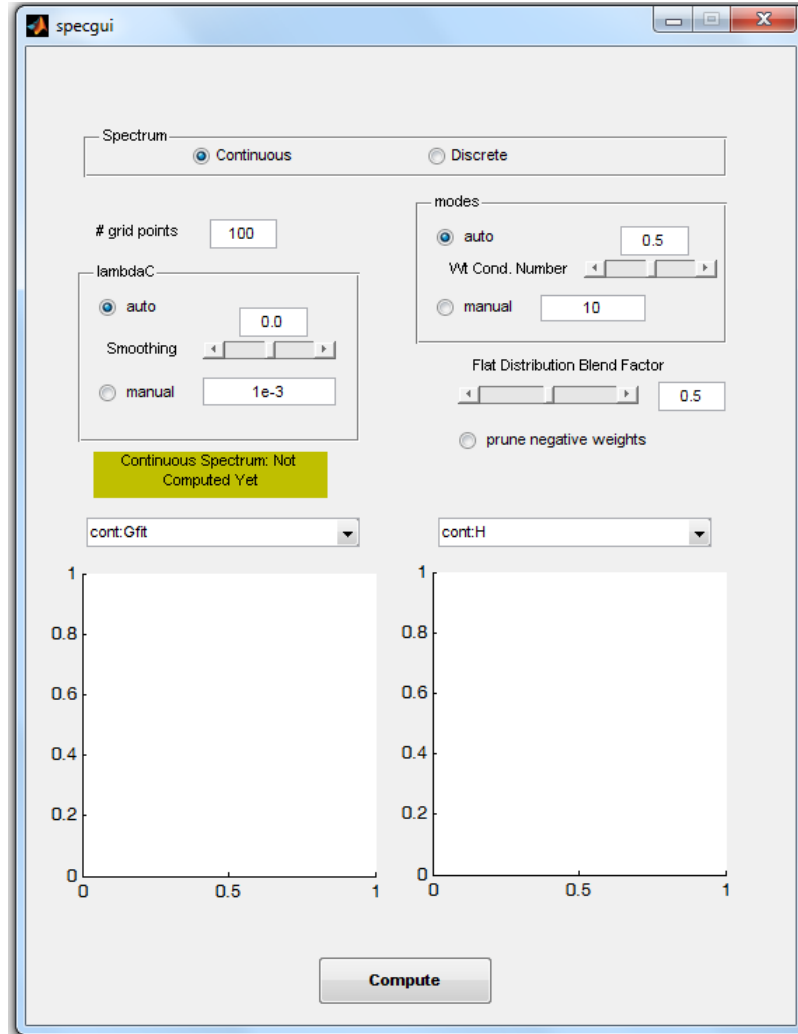


Figure 7: ReSpect GUI panel in Matlab

For the CRS, the parameters that can be adjusted are placed on the left-side of the console. These include the number of grid points for CRS, and λ_c , which may be adjusted manually or using the smoothing factor.

For the DRS, the parameters that can be adjusted are placed on the right-side of the console. These include the number and spacing of modes. The status of the CRS (whether computed or not) is prominently displayed on the middle left side of the panel.

There are two panels at the bottom of the console for plotting. One can use either or both of them for the CRS or DRS. Several plotting options (red box in figure 8) are included. Essentially they are plotting

interfaces to all the output files stored in output/.

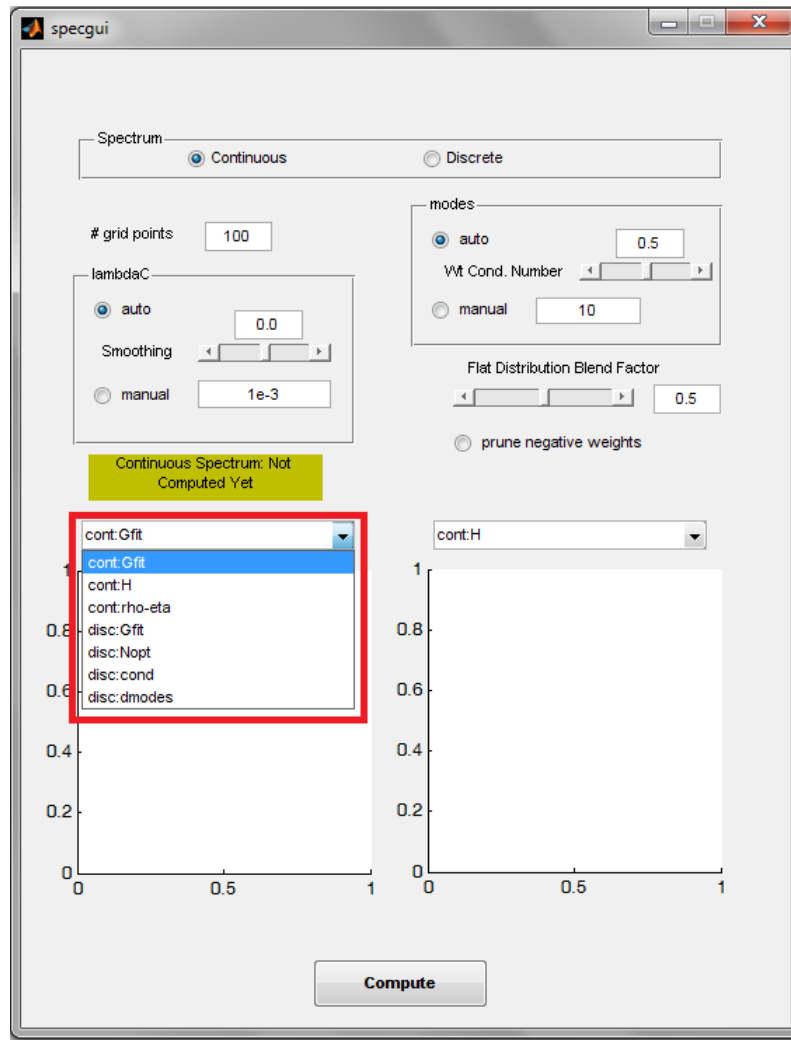


Figure 8: Plotting options (shown in the red box) for CRS and DRS.

Once the appropriate settings are selected, press the **Compute** button at the bottom of the GUI panel. ReSpect computes the relaxation spectrum, and plots the default charts.

Figure 9, for example, shows the continuous spectra and dynamic modulii calculated and plotted by GUI, respectively.

Notice also that the sign which previously said “Continuous Spectrum Not Computed Yet” has now changed to “Continuous Spectrum Computed”.

One can now choose the discrete radio button at the top of the panel, and press the **Compute** button, to compute the DRS.

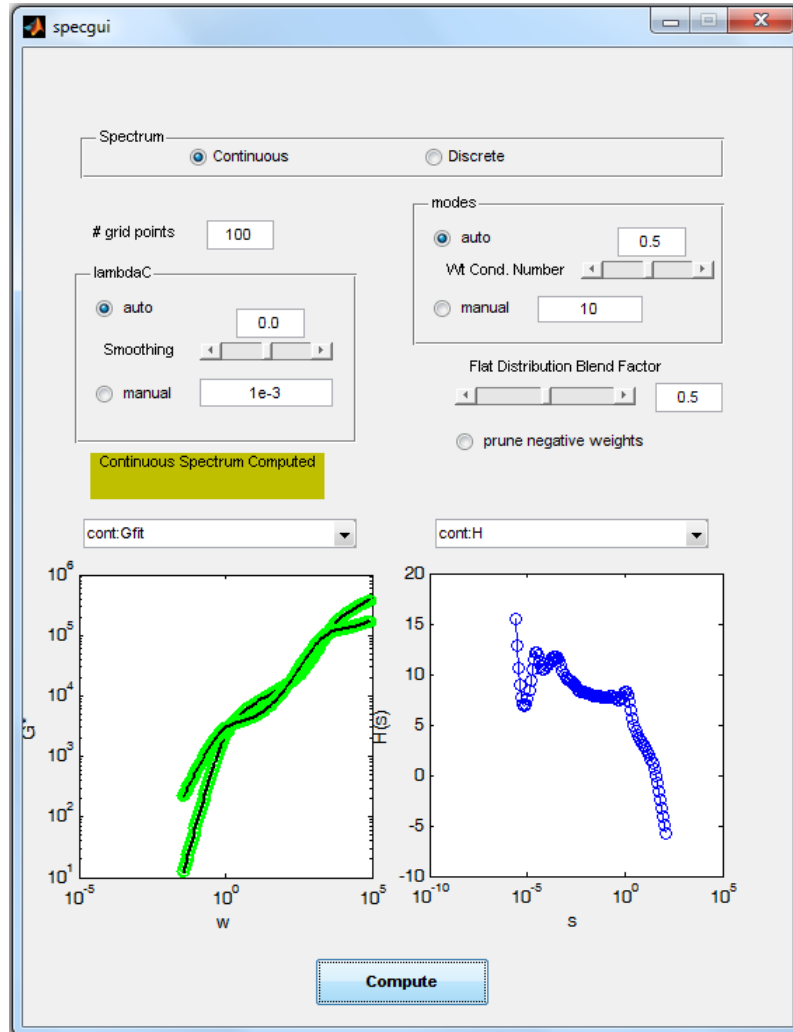


Figure 9: The left plot represents the dynamic modulus data (green symbols (lines) represent the experimental dynamic modulus (dynamic modulus from the corresponding CRS), respectively. Right box plot represents the extracted continuous relaxation spectrum.

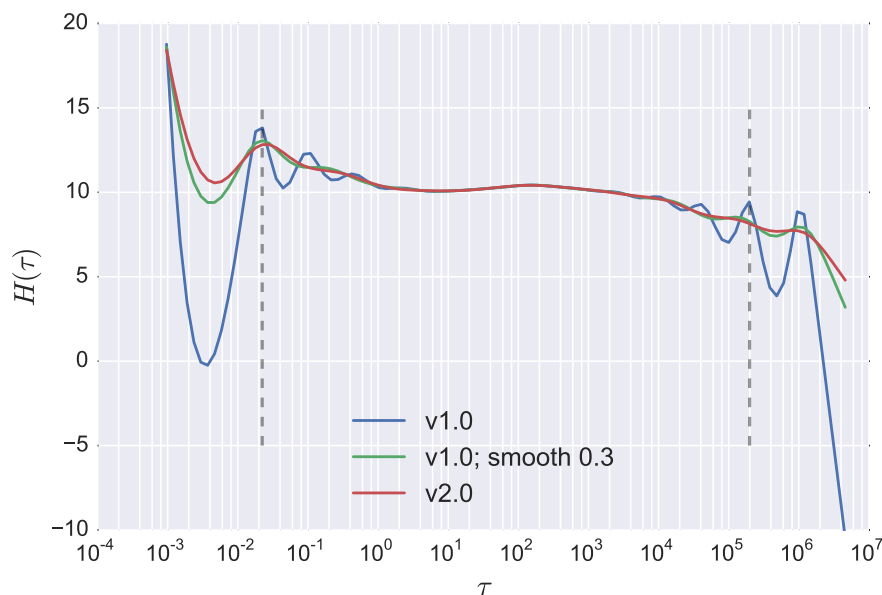


Figure 10: Spectrum extracted using v1.0 and v2.0 of ReSpect. The spectrum obtained from v1.0 with smoothing factor set to 0.3 is also shown. The dotted vertical lines demarcate the “trustable” portion of the spectrum (strict criterion).

5 FAQs

- (i) [The CRS inferred is too noisy or squiggly. How can I fix that?](#)

Response: Typically, the origin of this behavior is “noise” in the underlying $G^*(\omega)$. The program misinterprets the noise in the data as a valid signal, and tries to fit it, resulting in “squiggles”. An easy fix is to enhance smoothing by increasing the “smoothing factor” to a larger value.

This action skews the balance in eqn. 9 from ρ to η (which controls the smoothness of $H(\tau)$).

The default settings in the previous release of ReSpect (labeled v1.0, henceforth) made it especially susceptible. In the current release, ReSpect v2.0, the defaults are more sensible. In addition, some sanitization of the input data is carried out to suppress spurious squiggles.

To illustrate this, consider a dataset (courtesy of Prof. John Dealy) on H-polymers (sample H3A1). Figure 10 depicts the spectrum extracted using the original and current versions of ReSpect (v1.0, and v2.0, respectively). It is immediately obvious that the v1.0 spectrum has spurious oscillations especially near the ends of the trustable region. These oscillations can be suppressed by increasing the *smoothing factor*. In ReSpect, the default value of the smoothing factor is 0. To increase smoothing pick +ve values (less than one). In this example, we picked 0.3.

This is particularly evident, when one considers figure 11, which compares the experimental $G^*(\omega)$ for the H3A1 sample, with the $G^*(\omega)$ obtained from the three spectra shown in figure 10. The three

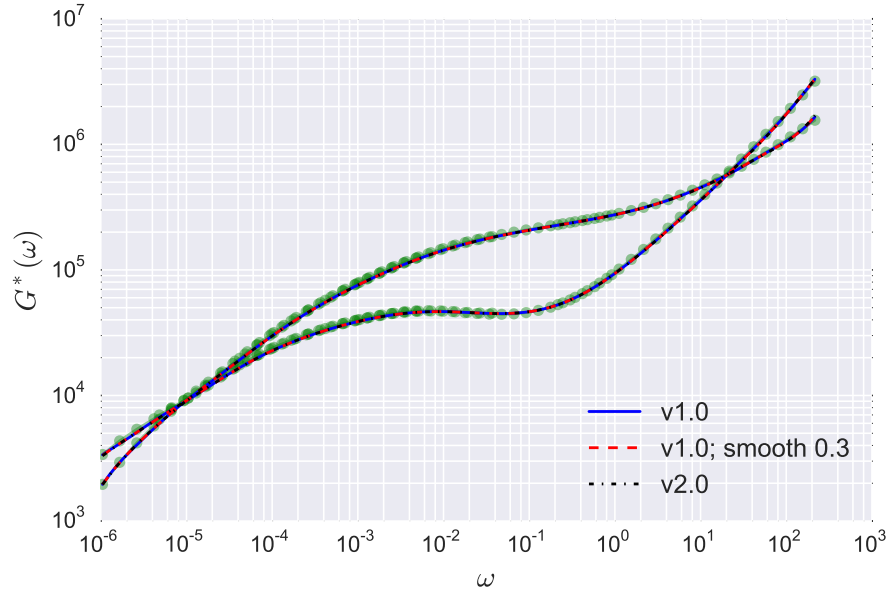


Figure 11: Comparison of the experimental $G^*(\omega)$ and that corresponding to the three different spectra extracted in figure 10.

$G^*(\omega)$ curves are practically indistinguishable.

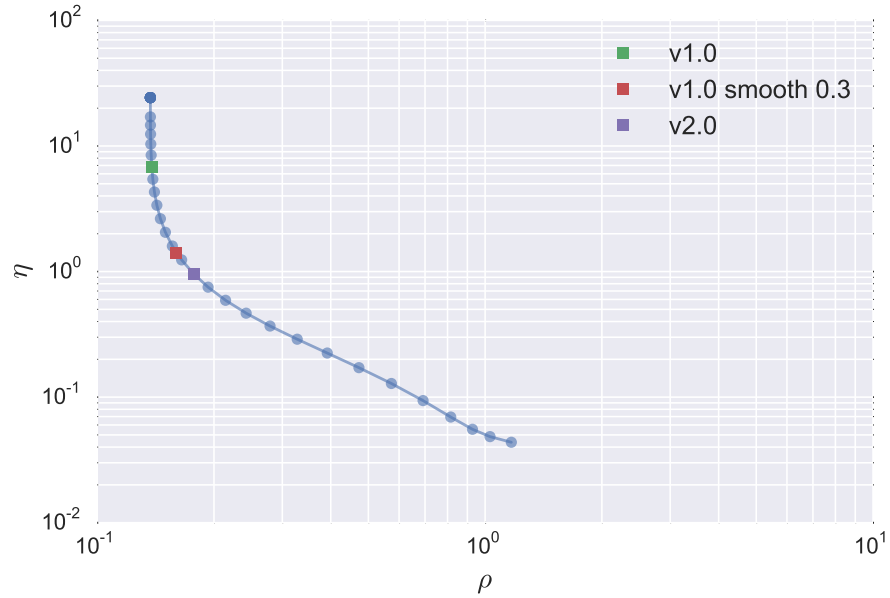


Figure 12: The $\rho - \eta$ curve, and the location corresponding to the critical λ_c for the three different spectra extracted in figure 10.

The relationship between the critical value of λ_c used in the three curves is made clear in the $\rho - \eta$ diagram in figure 12. In v1.0, $\lambda_c = 8.4 \times 10^{-6}$. When the smoothing factor was set to 0.3, $\lambda_c = 2.2 \times 10^{-3}$. In v2.0 of ReSpect, $\lambda_c = 10^{-2}$. One can observe the preference for greater smoothness

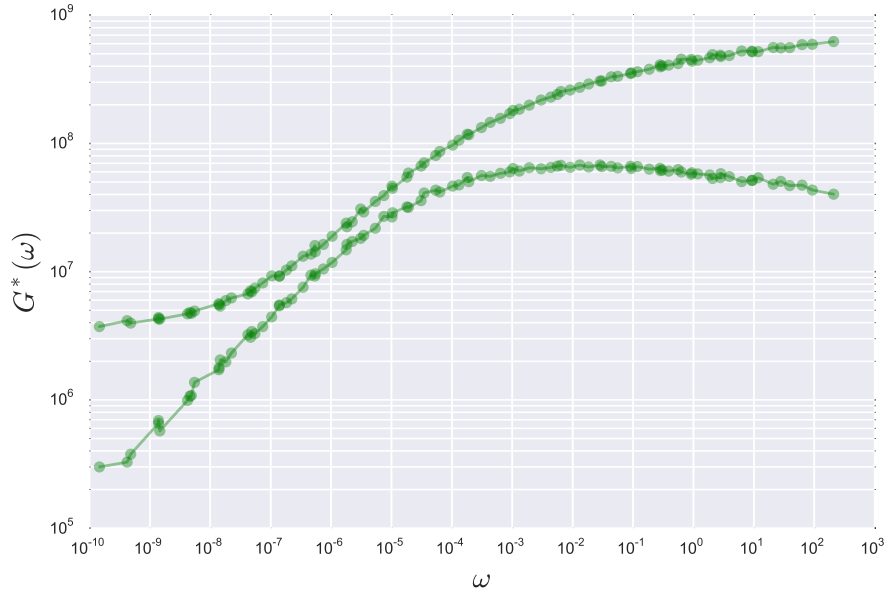


Figure 13: $G^*(\omega)$ with a plateau in $G'(\omega)$ at small ω .

with increasing λ_c in the $\rho - \eta$ diagram .

(ii) [How can I avoid -ve modes in the DRS?](#)

Response: First, make sure that the `par.prune` parameter in the `SetParameters` file is turned on. If you are using the GUI ensure, that the “prune negative weights” radio button is selected.

The DRS is inferred from the CRS. The smoother the CRS, the better the chance of avoiding -ve weights in the DRS. Compared to the prior release, ReSpect v2.0 is biased in the direction of greater smoothing of the CRS, and hence this problem should be less severe.

However, unlike the CRS, where we avoid -ve values of the spectrum by design (via $h(\tau) = \exp(H(\tau))$, which is positive, no matter what), one cannot guarantee a DRS with +ve values that fits the data adequately.

(iii) [My data has a plateau in \$G'\(\omega\)\$ at small \$\omega\$. Can I still use ReSpect to find the spectrum?](#)

Response: In some materials (for example, gels) the material might not relax completely, and there might be residual modulus at low frequencies. Thus the $G^*(\omega)$ might look like something in figure 13. ReSpect cannot directly find the spectrum for such $G^*(\omega)$.

This is because the actual spectrum for such materials (with a low-frequency plateau G_0 .) is related

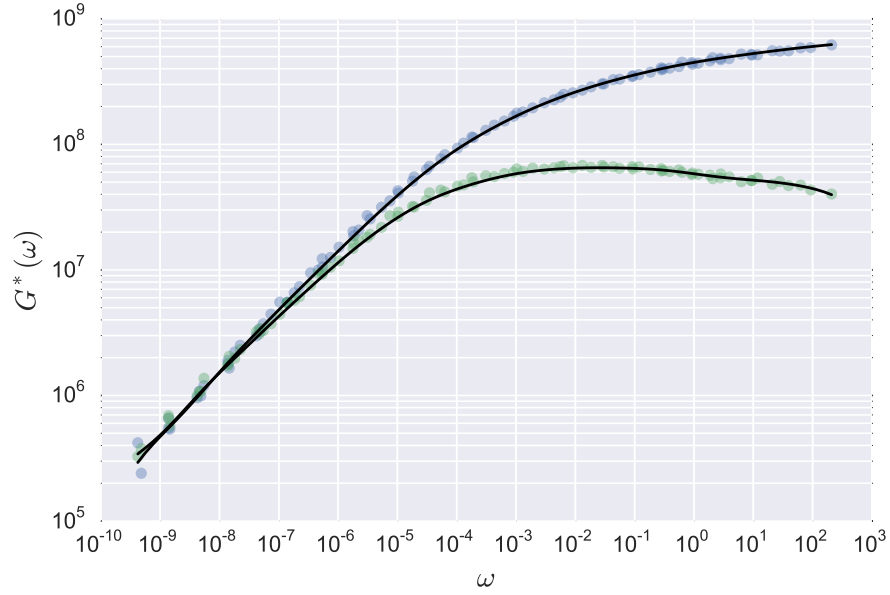


Figure 14: $G^*(\omega)$ with a plateau in $G'(\omega)$ at small ω . In this plot, the modified storage modulus $G'(\omega) - G_0$ was used in ReSpect to determine the spectrum. The lines show the $G^*(\omega)$ corresponding to the spectrum inferred in figure 15.

to $h(\tau)$ via a slightly different relation:

$$G'(\omega) = G_0 + \int_{-\infty}^{\infty} \frac{\omega^2 \tau^2}{1 + \omega^2 \tau^2} h(\tau) d\ln \tau = G_0 + \sum_{i=1}^N g_i \frac{\omega^2 \tau_i^2}{1 + \omega^2 \tau_i^2} \quad (4)$$

$$G''(\omega) = \int_{-\infty}^{\infty} \frac{\omega \tau}{1 + \omega^2 \tau^2} h(\tau) d\ln \tau = \sum_{i=1}^N g_i \frac{\omega \tau_i}{1 + \omega^2 \tau_i^2} \quad (5)$$

It can be easily remedied, by deleting the G_0 from the $G'(\omega)$ data (see figure 14). Note that redefining our storage modulus to $G'(\omega) - G_0$, makes these set of equations identical to the older ones, from which the spectrum can easily be extracted (figure 15).

(iv) [I can't find the CRS, no matter what I do! Help!](#)

Response: The first few things to try to fix are:

- are there any infinities or zeros in the input data?
- are there repeated frequencies?
- are you using enough number of points to resolve the CRS (increase to 200)
- **hard:** have you tried to look at the `InitializeH.m` function and modified it?

If your answer to most of these questions is yes, then perhaps your data does not obey the Kramer-Kronig relation between $G'(\omega)$ and $G''(\omega)$. This relation requires:

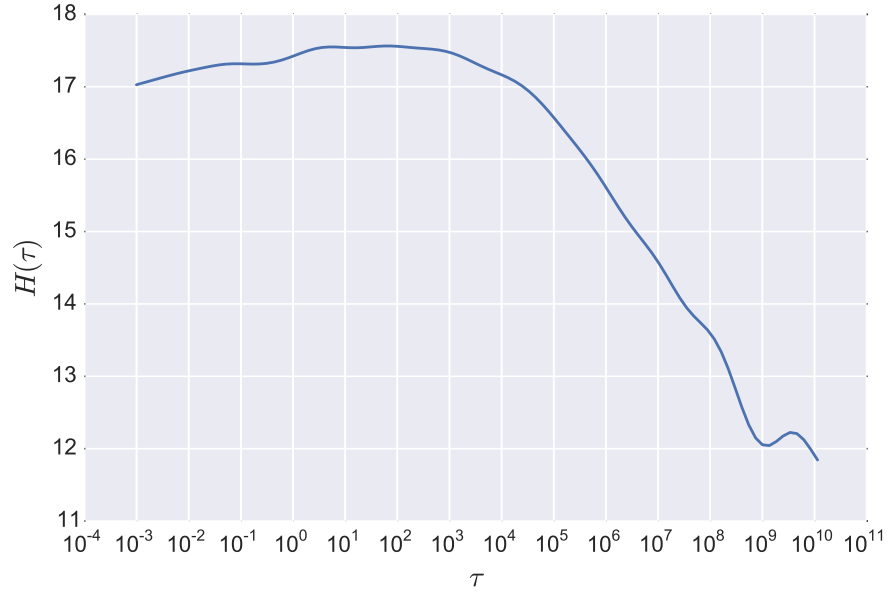


Figure 15: Spectrum corresponding to $G^*(\omega)$ with a plateau, after the value of the low-frequency plateau has been subtracted (see figure 14).

$$G'(\omega) = G_0 + \frac{2}{\pi} \int_0^\infty \frac{u G''(u)}{\omega^2 - u^2} du \quad (6)$$

$$G''(\omega) = \frac{2\omega}{\pi} \int_0^\infty \frac{G'(u)}{u^2 - \omega^2} du \quad (7)$$

If the inferred spectrum seems to either $G'(\omega)$ or $G''(\omega)$, but not the other, then it is a good sign that the Kramer-Kronig relation has been violated. By design, the $G^*(\omega)$ inferred from $H(\tau)$ obeys this constraint.

If you are still unable to find a spectrum, email me at sshanbhag@fsu.edu, with your input data and SetParameters file, and I will gladly take a look.

6 Program Organization

The directory `Gstar` contains the files with which the user interacts, namely, `SetParameters`, `contSpec`, `discSpec`, and `specgui`. In addition, the input data file (`Gst.dat` or `Gt.dat`) is in this directory.

There are five directories:

- (i) `common`: contains functions common to both DRS and CRS
- (ii) `continuous`: function used only by CRS
- (iii) `disc`: function used only by DRS
- (iv) `output`: all the calculated data files go into this subdirectory
- (v) `gui`: functions used by the GUI interface

The organization in of the `Gt` directory is completely analogous, except for the GUI interface.

6.1 Function Descriptions

1. User Facing Functions:

- `contSpec`
Computes CRS ($H(\tau)$) from a $G^*(\omega)$ using L-curve method.
- `discSpec`
Uses the CRS to determine an approximate DRS.
- `SetParameters`
Specify parameters used to extract the CRS or DRS
- `specgui`
provides the GUI interface .

2. Common Functions:

- `GetExpData`
Reads in experimental data for CRS and DRS calculation.
- `kernel`
Finds G' and G'' obtained from the corresponding inferred spectrum. Called from multiple files.
- `supsmu`
The supersmoother code from Mathworks. Used to smoothen input data in `GetExpData`.

3. Functions exclusively for CRS

- **InitializeH**
Guessed a spectrum to initialize CRS.
- **kernelD**
Approximates Jacobian of $V(\lambda)$ which is used by **LevenMarq** function.
- **lcurve**
Samples a range of λ , and calculates $H(\tau; \lambda)$ at each λ . Uses the L-curve method to estimate optimum λ . The method used to determine the optimal λ_c has been tweaked in v2.0. We define a function based on the normalized $\rho - \eta$ curve,

$$\epsilon(\lambda) = \frac{\rho(\lambda)}{\max \rho(\lambda)} + \frac{\eta(\lambda)}{\max \eta(\lambda)},$$

- **LevenMarq**
Finds the H for a given λ that minimizes the cost function.

$$\begin{aligned} V(\lambda) = & \frac{1}{n} \sum_{i=1}^n \left(\frac{(G'_e(\omega_i) - G'_i(\omega_i; H(\tau)))^2}{G'_e(\omega_i)} \right) + \\ & \frac{1}{n} \sum_{i=1}^n \left(\frac{(G''_e(\omega_i) - G''_i(\omega_i; H(\tau)))^2}{G''_e(\omega_i)} \right) + \\ & \lambda \int_{-\infty}^{\infty} \left(\frac{d^2}{d\tau^2} H(\tau) \right)^2 d \ln \tau \end{aligned} \quad (8)$$

4. Functions exclusively for DRS

- **GetWeights**
Finds the weight of each node by taking a weighted average of its contribution to G' and G'' (in DRS calculation).
- **GridDensity**
Takes in a PDF or density function, and spits out a bunch of points in accordance with the PDF.
- **MaxwellModes**
Solves the linear least-squares problem to obtain the DRS.
- **PlotMaxwellModes**
Plots experimental dynamic moduli and dynamic moduli obtained from the corresponding DRS.

5. Functions exclusively for GUI

- **specgui.fig**
Figure file for basic setup of GUI console.
- **updateplot**
Function called by **specgui** to update plots in the panel.